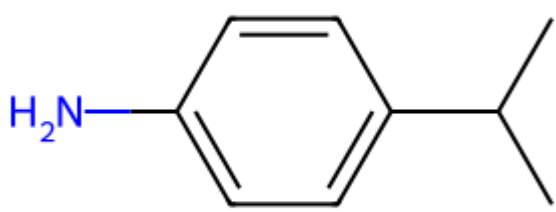
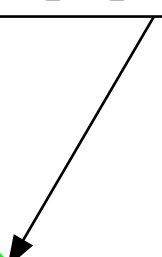


```
{'analysis_time': 0.0,  
'first_solution_time': 0.148014,  
'num_calls_rxn_model': 0,  
'num_calls_value_function': 0,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 9.57179816409135,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 9.57179816409135,  
'retro_star_value': 9.57179816409135}
```

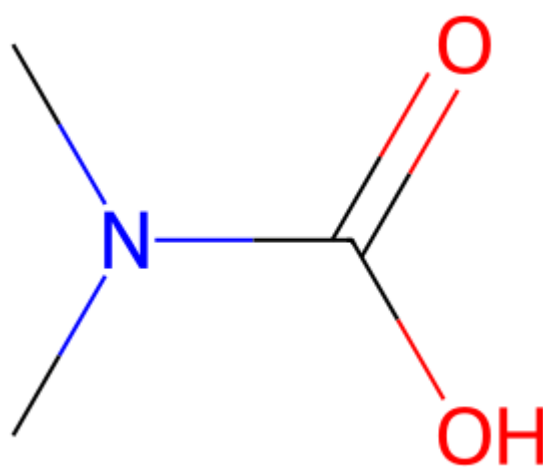


Reaction

```
{'analysis_time': 0.148014,  
'first_solution_time': 0.636708,  
'num_calls_rxn_model': 1,  
'num_calls_value_function': 0,  
'retro_star_min_cost': 10.177924398237888,  
'retro_star_reaction_number': 10.177924398237888,  
'retro_star_rxn_cost': 6.214608098422191,  
'retro_star_value': 10.177924398237888}
```



```
{'analysis_time': 0.148014,  
'first_solution_time': 0.148014,  
'num_calls_rxn_model': 1,  
'num_calls_value_function': 0,  
'retro_star_min_cost': 0.0,  
'retro_star_mol_cost': 0.0,  
'retro_star_reaction_number': 0.0,  
'retro_star_value': 10.177924398237888}
```

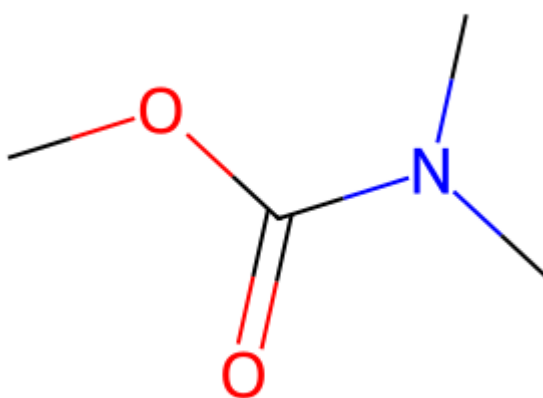


```
{'analysis_time': 0.148014,  
'first_solution_time': 0.636708,  
'num_calls_rxn_model': 1,  
'num_calls_value_function': 0,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 3.9633162998156966,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 3.9633162998156966,  
'retro_star_value': 10.177924398237888}
```



Reaction

```
{'analysis_time': 0.636708,  
'first_solution_time': 0.636708,  
'num_calls_rxn_model': 12,  
'num_calls_value_function': 396,  
'retro_star_min_cost': 5.115995809754082,  
'retro_star_reaction_number': 5.115995809754082,  
'retro_star_rxn_cost': 5.115995809754082,  
'retro_star_value': 11.330603908176274}
```



```
{'analysis_time': 0.636708,  
'first_solution_time': 0.636708,  
'num_calls_rxn_model': 12,  
'num_calls_value_function': 396,  
'retro_star_min_cost': 0.0,  
'retro_star_mol_cost': 0.0,  
'retro_star_reaction_number': 0.0,  
'retro_star_value': 11.330603908176274}
```